# Intelligent Vehicles Exercise 3 & 4 Report:

*Scan Matching, Sensor Fusion & Kalman Filter*


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**Abstract**

This report covers the implementation and evaluation of scan matching and sensor fusion techniques for localizing the Snowhite mobile robot. Dead reckoning from Exercise 2 provides continuous position estimates but suffers from unbounded drift. To correct this we implement the Cox scan matching algorithm which matches laser range finder readings against a known line-segment map using iterative least-squares minimization. Applying the Cox corrections directly (Task 4) keeps errors bounded within about 90 mm with a sawtooth pattern. Then we implement a Kalman filter to fuse dead reckoning with position fixes. In Task 5, the filter diverges because our process noise model underestimates heading uncertainty by roughly 3 times, causing the filter to become overconfident and ignore corrections. However, the Kalman filter with real Cox-based fixes (Task 6) works well, achieving the best performance at about 65 mm max error. This succeeds because the Cox correction is relative to the current estimate and its covariance is data-driven. The main lesson from this exercise is that Kalman filter requires realistic noise models to function properly and sensor fusion with well-characterized uncertainty outperforms either dead reckoning or scan matching alone.

# 1 Introduction

In Exercise 2 we implemented dead-reckoning estimation for the Snowhite robot and saw that the position uncertainty keeps growing over time without any bound. Dead reckoning works well for short distances but accumulated errors in the wheel encoders and steering angle cause the estimated trajectory to drift further and further from the true path. To correct the drift we use sensor fusion. We combine the relative position estimates from dead reckoning with absolute position observations from an external sensor. Here the Snowhite robot carries a laser range finder (LIDAR) and the environment is modeled as a set of known line segments (walls). By matching what the laser sees against this known map we can compute a position correction a get a position fix.

The Cox scan matching algorithm [1] handles this correction. It iteratively adjusts the robot's believed position until the laser measurements best align with the map lines using least-squares. The Kalman filter [2] then gives us the optimal way to fuse the dead-reckoning prediction with the scan-matching observation weighting each source by how confident we are in it (the Kalman Gain).

This report covers the theoretical background (Section 2), including the dead-reckoning model, the Cox algorithm, coordinate transformations and the Kalman filter equations. Section 3 presents the experiments and results for Tasks 2, 4, 5, and 6. Finally, Section 4 wraps up with conclusions and suggestions for improvement.

# 2 Theory / Method

## 2.1 Dead Reckoning Model

The Snowhite robot uses a tricycle kinematic model with wheelbase $L = 680$ mm and sampling period $T = 0.050$ s. The state vector is $\mathbf{X} = [x, y, \theta]^T$, where $(x, y)$ is position and $\theta$ is heading. Given velocity $v$ and steering angle $\alpha$, the state update is:

$$\Delta D = v \cos(\alpha) \cdot T \tag{1}$$

$$\Delta\theta = \frac{v \sin(\alpha) \cdot T}{L} \tag{2}$$

$$x_k = x_{k-1} + \Delta D \cos\left(\theta_{k-1} + \frac{\Delta\theta}{2}\right) \tag{3}$$

$$y_k = y_{k-1} + \Delta D \sin\left(\theta_{k-1} + \frac{\Delta\theta}{2}\right) \tag{4}$$

$$\theta_k = \theta_{k-1} + \Delta\theta \tag{5}$$

## 2.2 Error Propagation

The covariance matrix $\Sigma_k$ is propagated at each timestep using:

$$\Sigma_k = J_X \, \Sigma_{k-1} \, J_X^T + J_U \, \Sigma_U \, J_U^T \tag{6}$$

where $J_X$ is the Jacobian with respect to the state and $J_U$ is the Jacobian with respect to the control inputs.

The state Jacobian using the midpoint angle $\theta_m = \theta_{k-1} + \Delta\theta/2$ is:

$$J_X = \begin{bmatrix} 1 & 0 & -\Delta D \sin(\theta_m) \\ 0 & 1 & \Delta D \cos(\theta_m) \\ 0 & 0 & 1 \end{bmatrix} \tag{7}$$

For the input Jacobian $J_U$ with respect to $[v, \alpha, T]$ we use the shorthand $c_\theta = \cos\theta_m$, $s_\theta = \sin\theta_m$, $c_\alpha = \cos\alpha$, $s_\alpha = \sin\alpha$. Applying the chain rule to Equations (1) through (5) (keeping in mind that $\theta_m$ itself depends on $v$, $\alpha$, and $T$), we get:

$$J_U = \begin{bmatrix} Tc_\alpha c_\theta - \dfrac{vc_\alpha s_\alpha T^2 s_\theta}{2L} & -vs_\alpha T c_\theta - \dfrac{v^2 c_\alpha^2 T^2 s_\theta}{2L} & vc_\alpha c_\theta - \dfrac{v^2 c_\alpha s_\alpha T s_\theta}{2L} \\ Tc_\alpha s_\theta + \dfrac{vc_\alpha s_\alpha T^2 c_\theta}{2L} & -vs_\alpha T s_\theta + \dfrac{v^2 c_\alpha^2 T^2 c_\theta}{2L} & vc_\alpha s_\theta + \dfrac{v^2 c_\alpha s_\alpha T c_\theta}{2L} \\ \dfrac{s_\alpha T}{L} & \dfrac{vc_\alpha T}{L} & \dfrac{vs_\alpha}{L} \end{bmatrix} \tag{8}$$

Each entry comes from differentiating $x_k = x_{k-1} + \Delta D \cos\theta_m$, $y_k = y_{k-1} + \Delta D \sin\theta_m$, and $\theta_k = \theta_{k-1} + \Delta\theta$ using the product rule, since both $\Delta D$ and $\theta_m$ depend on $v$, $\alpha$, and $T$. As an example:

$$\frac{\partial x_k}{\partial v} = \underbrace{\frac{\partial \Delta D}{\partial v}}_{c_\alpha T} \cos\theta_m \; - \; \Delta D \sin\theta_m \underbrace{\frac{\partial \theta_m}{\partial v}}_{\frac{s_\alpha T}{2L}}$$

The input covariance matrix is:

$$\Sigma_U = \begin{bmatrix} \sigma_v^2 & 0 & 0 \\ 0 & \sigma_\alpha^2 & 0 \\ 0 & 0 & \sigma_T^2 \end{bmatrix} \tag{9}$$

We use a proportional plus floor noise model: $\sigma_v = 0.05|v| + 10$ mm/s, $\sigma_\alpha = 0.02|\alpha| + 0.5°$, and $\sigma_T = 0.001$ s. This way the uncertainty scales with how fast the robot is moving while never dropping to zero even when the robot is stationary or going straight.

## 2.3 Coordinate Frame Transformations

The laser is mounted on the robot with offset parameters: $\alpha_s = 660$ mm forward, $\beta_s = 0$ mm lateral, and $\gamma_s = -\pi/2$ rad angular offset. To go from a raw laser reading to a world coordinate, we need three transformations.

First polar to Cartesian in the sensor frame:

$$x_s = r\cos(\phi), \quad y_s = r\sin(\phi) \tag{10}$$

where $r$ is the measured distance and $\phi$ is the beam angle.

Then sensor frame to robot frame:

$$\begin{bmatrix} x_r \\ y_r \end{bmatrix} = \begin{bmatrix} \cos\gamma_s & -\sin\gamma_s \\ \sin\gamma_s & \cos\gamma_s \end{bmatrix} \begin{bmatrix} x_s \\ y_s \end{bmatrix} + \begin{bmatrix} \alpha_s \\ \beta_s \end{bmatrix} \tag{11}$$

And finally robot frame to world frame:

$$\begin{bmatrix} x_w \\ y_w \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x_r \\ y_r \end{bmatrix} + \begin{bmatrix} R_x \\ R_y \end{bmatrix} \tag{12}$$

where $(R_x, R_y, \theta)$ is the current believed robot pose.

## 2.4 Cox Scan Matching Algorithm

The Cox algorithm iteratively finds a position correction $\mathbf{b} = [dx, dy, d\theta]^T$ that minimizes the distances from laser points to the nearest map line segments. It works in five steps.

**Step 0:** For each line segment $L_i$ from point $P_1$ to $P_2$, compute the unit normal vector and perpendicular distance from the origin:

$$\mathbf{u}_i = \frac{1}{\|\mathbf{d}\|} \begin{bmatrix} -d_y \\ d_x \end{bmatrix}, \quad r_i = \mathbf{u}_i \cdot \mathbf{z}_i \tag{13}$$

where $\mathbf{d} = P_2 - P_1$ is the direction along the line and $\mathbf{z}_i$ is any point on it.

**Step 1:** Transform all laser measurements from sensor to world coordinates using Equations (10) through (12), with the pose updated by the accumulated correction $[ddx, ddy, dd\theta]$ from previous iterations.

**Step 2:** For each transformed point $\mathbf{v}_j$, compute the signed perpendicular distance to every line:

$$y_j = r_i - \mathbf{u}_i \cdot \mathbf{v}_j \tag{14}$$

Assign $\mathbf{v}_j$ to whichever line $L_i$ minimizes $|y_j|$. Any point with $|y_j| > 200$ mm is thrown out as an outlier.

**Step 3:** For each valid point $j$ assigned to line $i$, build one row of the design matrix:

$$A_j = \begin{bmatrix} u_{i,1} & u_{i,2} & \mathbf{u}_i \cdot \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} (\mathbf{v}_j - \mathbf{v}_m) \end{bmatrix} \tag{15}$$

where $\mathbf{v}_m = [R_x + ddx, R_y + ddy]^T$ is the current robot position estimate. Then solve by least squares:

$$\mathbf{b} = (A^T A)^{-1} A^T \mathbf{Y} \tag{16}$$

The residual variance and the covariance of the correction are:

$$S^2 = \frac{(\mathbf{Y} - A\mathbf{b})^T (\mathbf{Y} - A\mathbf{b})}{n - 4}, \quad C = S^2 (A^T A)^{-1} \tag{17}$$

**Steps 4 and 5:** Accumulate the correction ($ddx \mathrel{+}= dx$, $ddy \mathrel{+}= dy$, $dd\theta \mathrel{+}= d\theta$) and check convergence. If $\sqrt{dx^2 + dy^2} < 5$ mm and $|d\theta| < 0.1°$, stop. Otherwise go back to Step 1.

## 2.5 Kalman Filter

The Kalman filter fuses the dead-reckoning estimate $\hat{X}^-(i)$ (with covariance $\Sigma_X(i)$) and a position fix $\hat{X}_{pf}(i)$ (with covariance $\Sigma_{pf}(i)$). The updated state estimate is:

$$\hat{X}^+(i) = \Sigma_{pf}(i) \left(\Sigma_{pf}(i) + \Sigma_X(i)\right)^{-1} \hat{X}^-(i) + \Sigma_X(i) \left(\Sigma_{pf}(i) + \Sigma_X(i)\right)^{-1} \hat{X}_{pf}(i) \tag{18}$$

and the updated covariance is:

$$\Sigma_X^+(i) = \left(\Sigma_{pf}^{-1}(i) + \Sigma_X^{-1}(i)\right)^{-1} \tag{19}$$

This is essentially a weighted average where the weights are inversely proportional to each estimate's uncertainty. The result is always more certain than either source alone.

For Task 5 (simulated position fixes), we generate the fix by adding zero-mean Gaussian noise to the ground truth:

$$\hat{X}_{pf} = \begin{bmatrix} x_{true} + \varepsilon_x \\ y_{true} + \varepsilon_y \\ \theta_{true} + \varepsilon_\theta \end{bmatrix}, \quad \Sigma_{pf} = \begin{bmatrix} \sigma_x^2 & 0 & 0 \\ 0 & \sigma_y^2 & 0 \\ 0 & 0 & \sigma_\theta^2 \end{bmatrix} \tag{20}$$

For Task 6 (Cox-based fixes), the position fix is the current dead-reckoning position corrected by the Cox output:

$$\hat{X}_{pf} = \begin{bmatrix} X(k) + dx \\ Y(k) + dy \\ A(k) + d\theta \end{bmatrix}, \quad \Sigma_{pf} = C_{\text{Cox}} \tag{21}$$

# 3 Experiments and Results

The experiments use data collected by the Snowhite robot navigating an indoor environment modeled by 23 reference nodes and 16 line segments. The control data has 4050 timesteps of velocity and steering angle measurements at $T = 50$ ms intervals, with ground truth positions from an external tracking system. There are 47 laser scans available at specific timesteps during the run.

## 3.1 Task 2: Dead Reckoning with Uncertainty

**Task 2**: Complete the script to predict the uncertainty of the dead reckoning position.

We implemented the covariance propagation from Equation (6) in `main.m`. At each timestep, both Jacobians $J_X$ and $J_U$ are computed and the covariance matrix is propagated forward. We store it as a flattened $1 \times 9$ vector per timestep.

For noise we use the proportional plus floor model described in Section 2.2: $\sigma_v = 0.05|v| + 10$ mm/s, $\sigma_\alpha = 0.02|\alpha| + 0.5°$, and $\sigma_T = 0.001$ s. The floor terms make sure uncertainty never drops to zero even when the robot is stationary or driving straight.

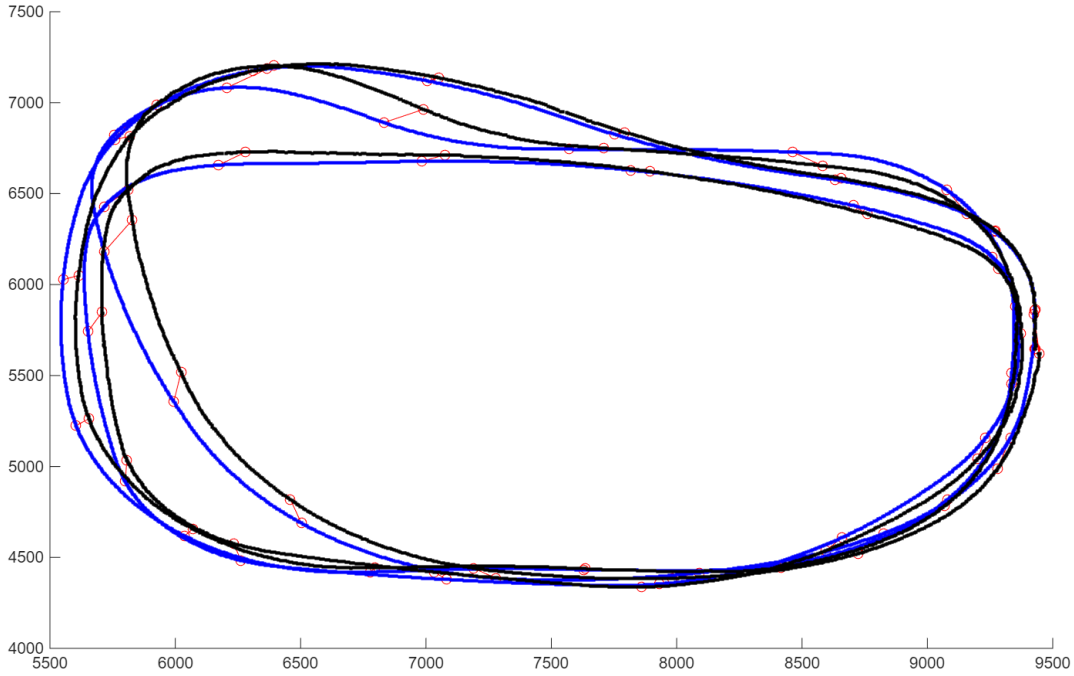


Figure 1: Dead reckoning path (blue) vs. ground truth (black) form Task 2.

Red circles mark laser scan positions. The robot does roughly three loops. In the first loop the blue and black paths nearly overlap, but by the second and third loops a clear lateral offset develops, especially on the left and top side of the trajectory.

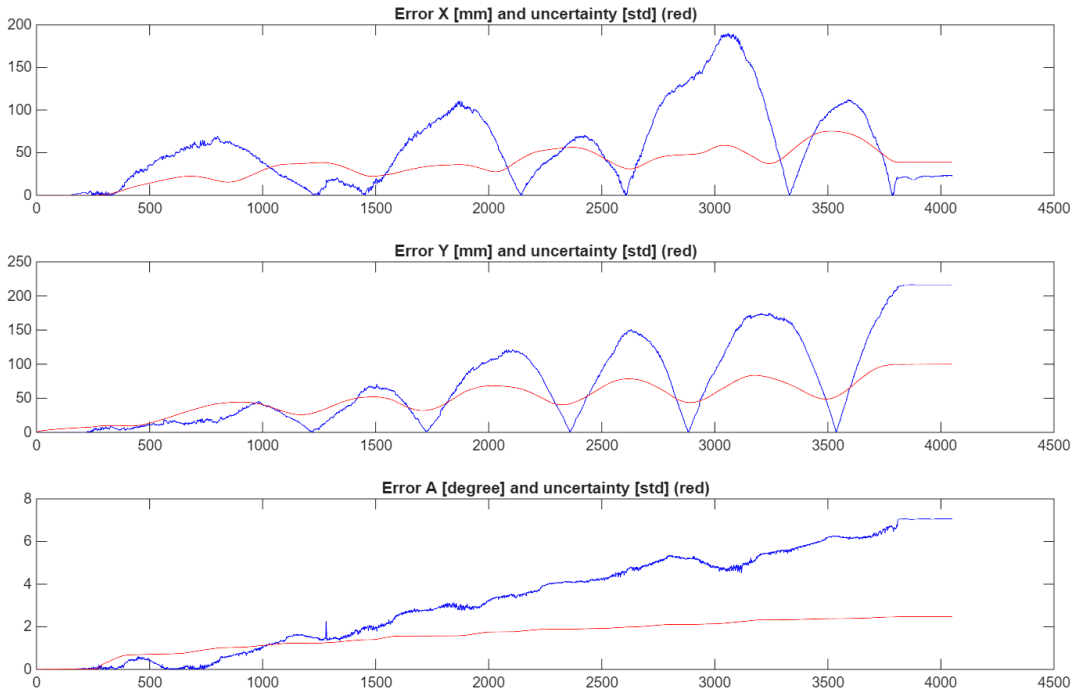


Figure 2: Absolute error (blue) and estimated $1\sigma$ uncertainty (red) form Task 2.

$X$ error grows to about 190 mm, $Y$ error reaches about 210 mm and heading error increases steadily to roughly 7°.

The results are what we expected. The position error keeps growing (Figure 2), reaching about 190 mm in $X$ and 210 mm in $Y$ by the end of the run. The heading error keep growing as it accumulates to about 7°, and since heading error directly amplifies future position errors through the cosine and sine terms in the kinematic model it creates the accelerating drift visible in Figure 1.

The red $1\sigma$ bounds from our covariance propagation grow smoothly reaching about 70 mm for $X$, 90 mm for $Y$ and 2.5° for heading. However, the actual error frequently exceeds these bounds particularly for heading where the true error is roughly $3\times$ the estimated $1\sigma$. This tells us our process noise model is a bit optimistic. The noise floors ($\sigma_v = 10$ mm/s, $\sigma_\alpha = 0.5°$) do not fully capture the systematic errors in the real encoder and steering data.

## 3.2 Task 4: Dead Reckoning And Cox Correction

**Task 4**: Call Cox_LineFit() from the main script and update position estimates using the match results (dx, dy, da and C). Plot the errors and estimated error covariances along the path.

We implemented the Cox scan matching algorithm in `Cox_LineFit.m` following the steps in Section 2.4. At each laser scan the algorithm runs and the returned correction $(dx, dy, d\theta)$ is applied directly to the robot's position. The covariance is reset to the Cox uncertainty $C$ after each correction.

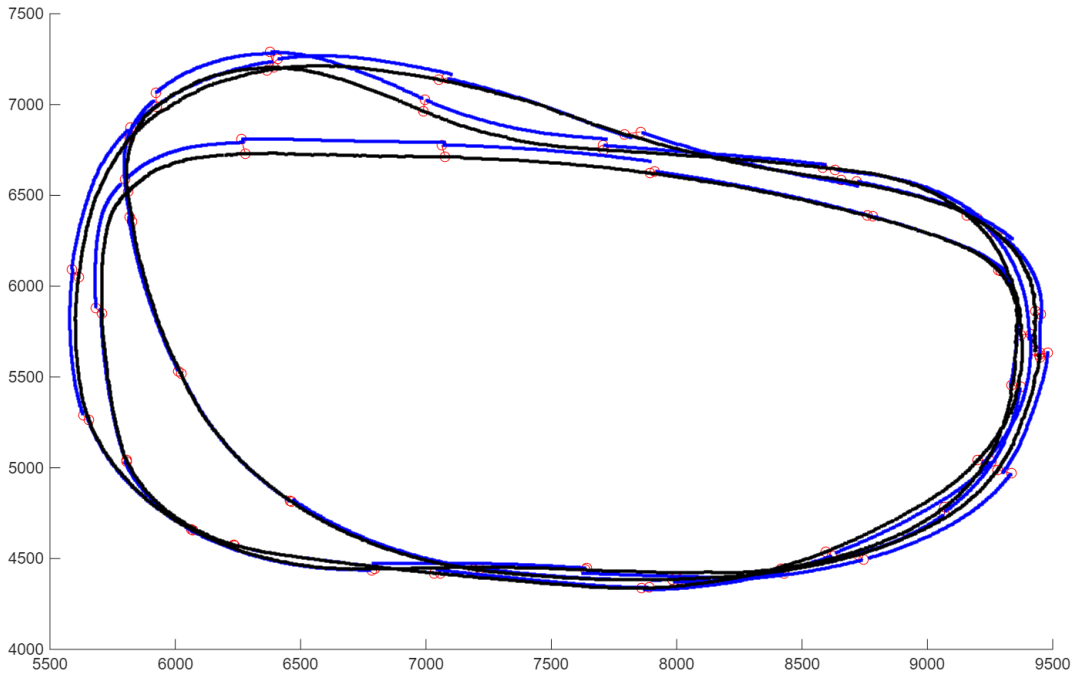


Figure 3: Trajectory with Cox corrections (Task 4).

The blue line is the estimated path and the black is the ground truth. Red circles are scan positions. Compared to Figure 1, the blue path tracks the ground truth much more closely through all three loops.

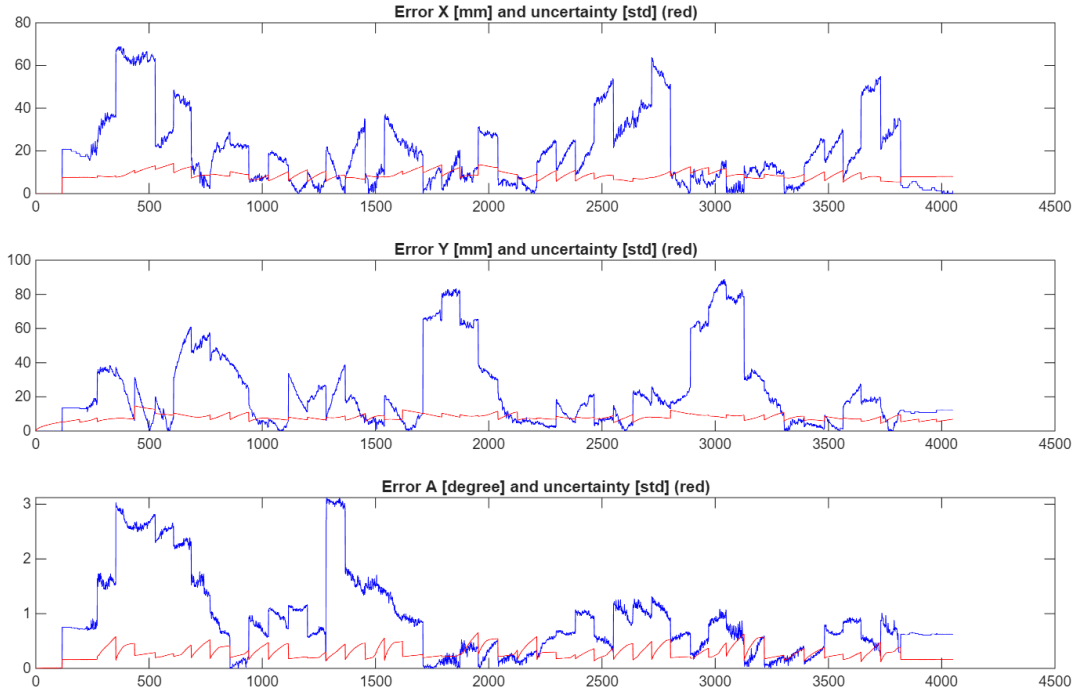


Figure 4: Error and uncertainty form Task 4.

The $X$ error stays within about 70 mm, $Y$ within about 90 mm and heading within about 3° all showing the characteristic sawtooth pattern of periodic corrections. The red $1\sigma$ line resets at each scan when the covariance is replaced by the Cox uncertainty $C$. The Cox algorithm matches the laser readings to the map and pulls the position back at each scan. In the error plot (Figure 4) you can clearly see the sawtooth pattern as between scans the error grows from dead-reckoning drift (rising edges), and at each scan the Cox correction

brings it sharply back down (falling edges). The peak errors of about 70 mm in $X$ and 90 mm in $Y$ are much smaller than the 190 to 210 mm we saw in Task 2.

The red $1\sigma$ line shows step-like resets because the covariance gets directly replaced by the Cox covariance $C$ after each correction. Between corrections, uncertainty grows via dead-reckoning propagation. The actual error still exceeds the $1\sigma$ estimate especially in heading which is consistent with what we saw in Task 2. One downside of this approach is that the corrections can be abrupt, since the Cox result is applied directly without weighing it against how confident the dead reckoning was.

### 3.3 Task 5: Kalman Filter with Simulated Fixes

**Task 5**: Simulate position fixes by adding Gaussian noise to ground truth positions. Use a Kalman filter to fuse the simulated fixes with dead reckoning. Run with (a) $\sigma_x = \sigma_y = 10$ mm, $\sigma_\theta = 1°$ and (b) $\sigma_x = \sigma_y = 100$ mm, $\sigma_\theta = 3°$.

We implemented the Kalman filter update from Equations (18) and (19) in the `TASK == 51` and `TASK == 52` blocks. Simulated position fixes are generated by adding Gaussian noise to the ground truth at timestamps where laser scans occur.

**Task 5a** ($\sigma_x = \sigma_y = 10$ mm, $\sigma_\theta = 1°$):

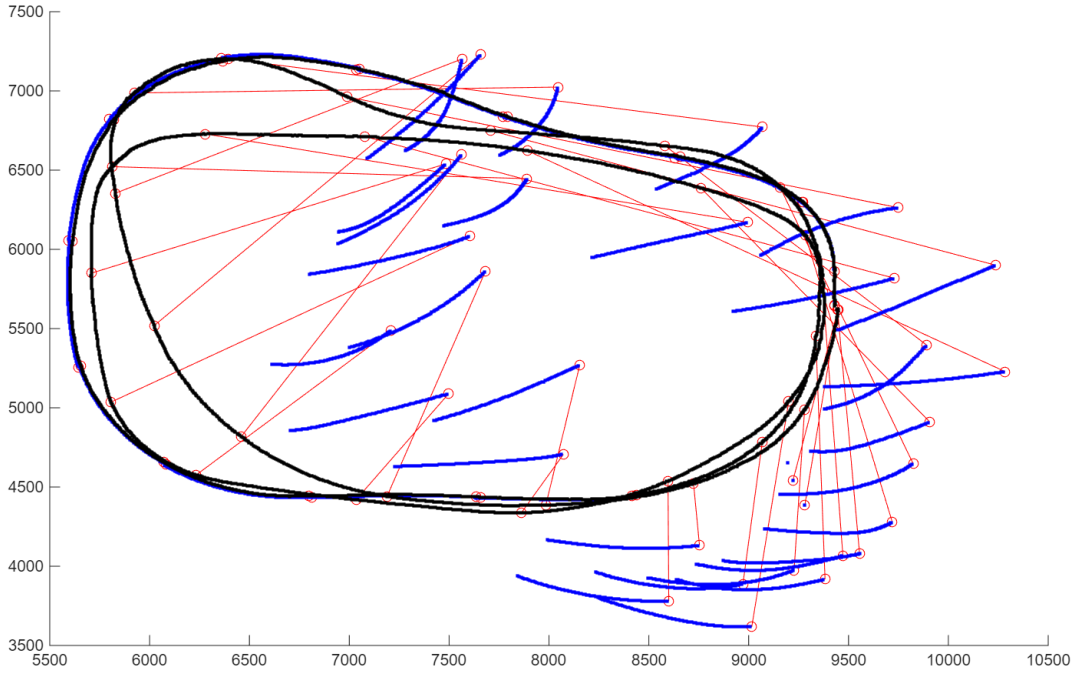


Figure 5: Task 5a path.

The filter tracks well during the first loop, but then diverges catastrophically. Blue segments scatter far from the true path (black) with red correction lines spanning thousands of millimeters.

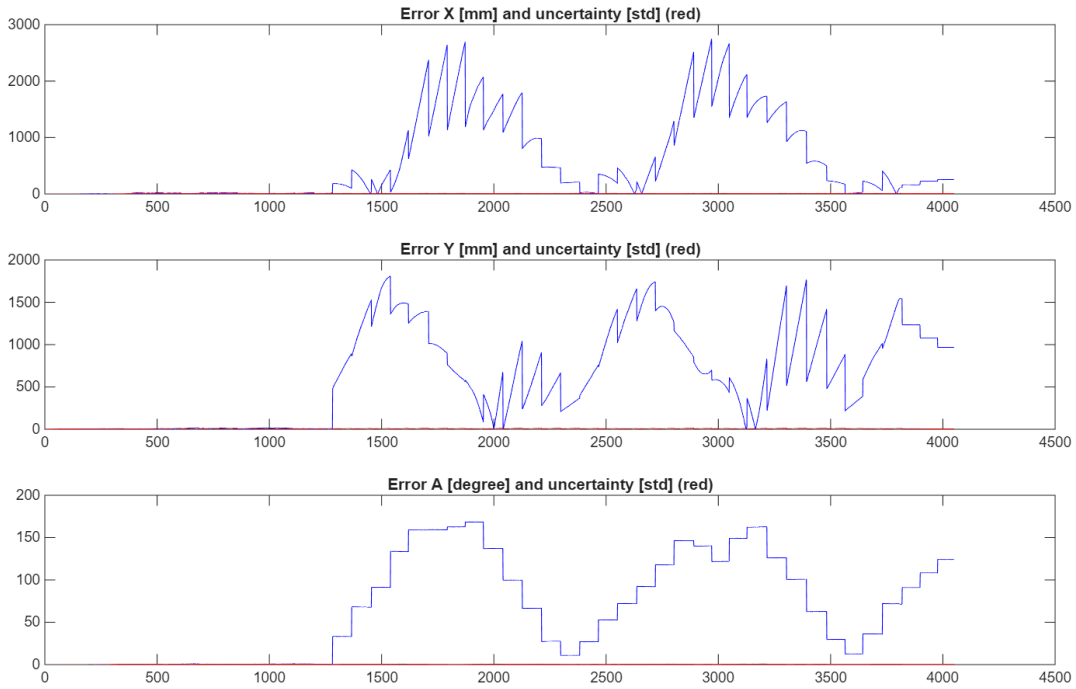


Figure 6: Task 5a errors.

$X$ error reaches about 3000 mm, $Y$ about 1800 mm, and heading about 160°. The red $1\sigma$ line is flat near zero meaning the filter thinks it knows the position to within a few millimeters while the actual error is in the meters.

This was a surprising result. We expected the small-noise fixes to give excellent tracking but instead the Kalman filter diverges (Figure 5). It works fine for the first portion of the run then completely loses the true trajectory.

The reason we think is our process noise model underestimates the true heading uncertainty. Here is what happens step by step. After a simulated fix with $\sigma_\theta = 1°$ the Kalman filter collapses the heading covariance to roughly $(1°)^2$. During the next dead-reckoning interval the modeled heading uncertainty grows too slowly because our noise parameters are too optimistic. So when the next fix arrives, the Kalman gain for heading is small because the filter still thinks its heading estimate is good. The heading correction barely gets applied. This creates a vicious cycle and the heading error accumulates uncorrected which causes large position errors through the kinematic model and the filter's reported covariance (the red line near zero in Figure 6) becomes completely disconnected from reality. This is a case of Kalman filter inconsistency from an overly optimistic process noise model.

**Task 5b** ($\sigma_x = \sigma_y = 100$ mm, $\sigma_\theta = 3°$):

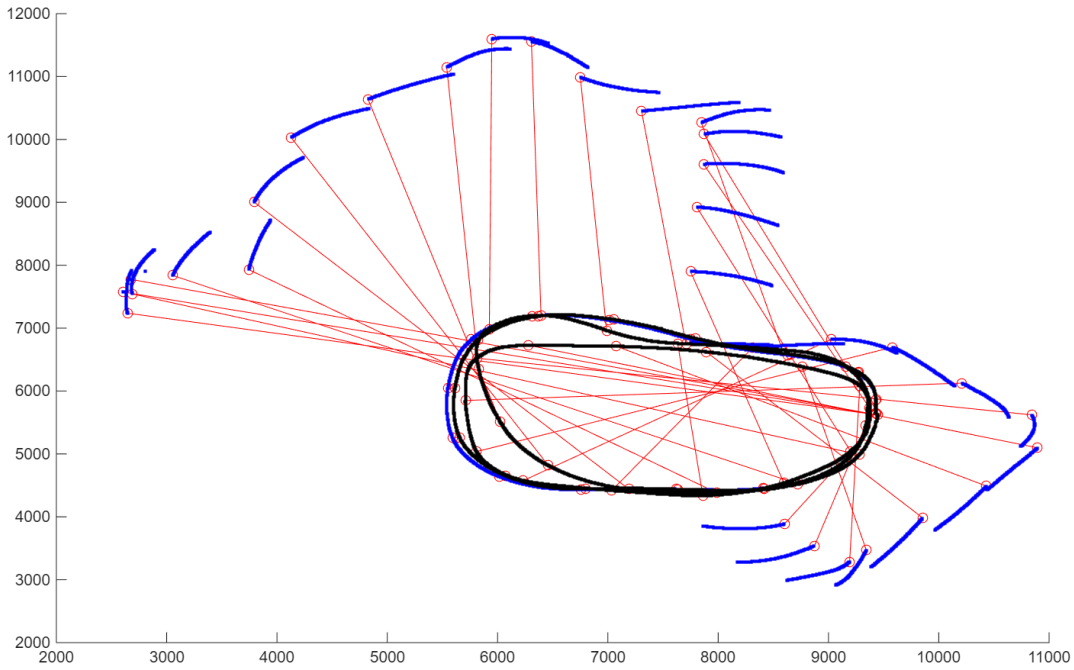


Figure 7: Task 5b path.

Even worse than 5a. The blue trajectory spirals far outside the operating area, reaching coordinates up to $(11000, 12000)$ mm while the true path stays within roughly (5500–9500, 4000–7500) mm. The ground truth loops are barely visible in the lower center.

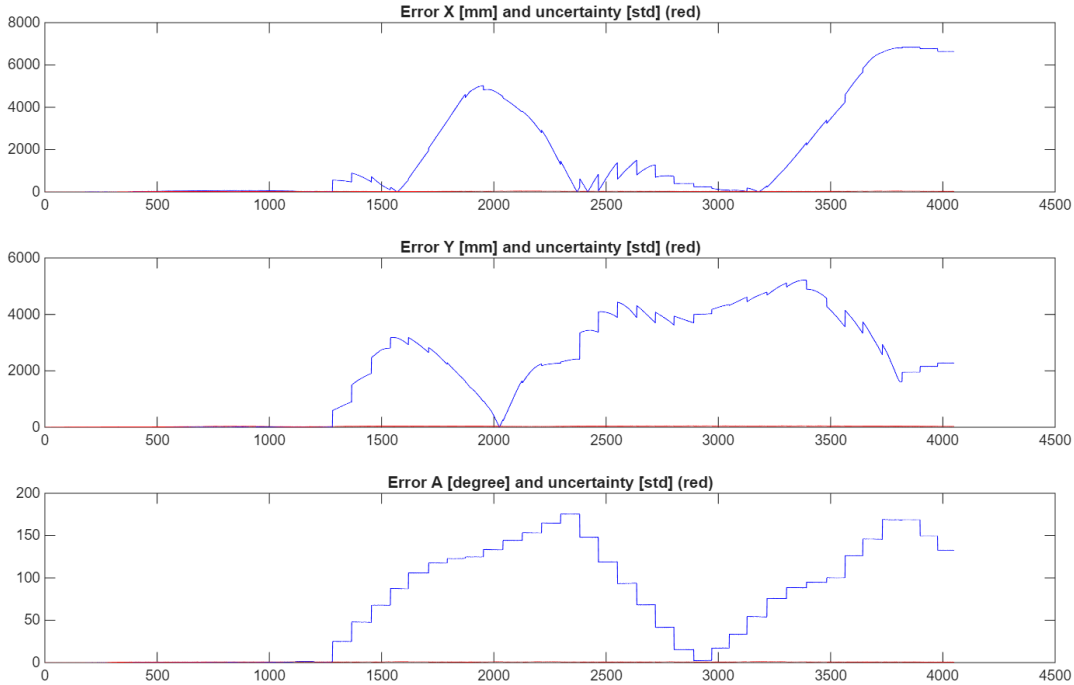


Figure 8: Task 5b errors.

$X$ error reaches about 7000 mm, $Y$ about 5000 mm, heading about 170°. Same overconfidence problem as 5a, but with even larger actual errors.

Task 5b shows the same divergence as 5a but worse (Figure 7). With the larger fix noise ($\sigma = 100$ mm, 3°) the Kalman gain at each fix is even smaller since the filter trusts the noisier fix less. So the heading corrections get applied even more weakly. The $X$ error reaches about 7000 mm and the heading error approaches 170°

(Figure 8) meaning the robot essentially believes it is facing the opposite direction.

Both 5a and 5b diverge for the same reason which is that the process noise model underestimates heading uncertainty making the filter overconfident. Task 5b is worse because the larger fix uncertainty makes the filter trust corrections even less. The practical lesson here is that a Kalman filter is only as good as its noise models. When the process noise is underestimated the filter becomes overconfident and ignores corrections and diverges.

### 3.4 Task 6: Kalman Filter with Cox Fixes

**Task 6**: Use a Kalman filter to fuse the position estimate from dead reckoning with position estimates from the Cox scan matching algorithm.

Here we replace the simulated fixes with real Cox scan matching results. The position fix is the current dead-reckoning position plus the Cox correction with the Cox covariance $C$ as $\Sigma_{pf}$.

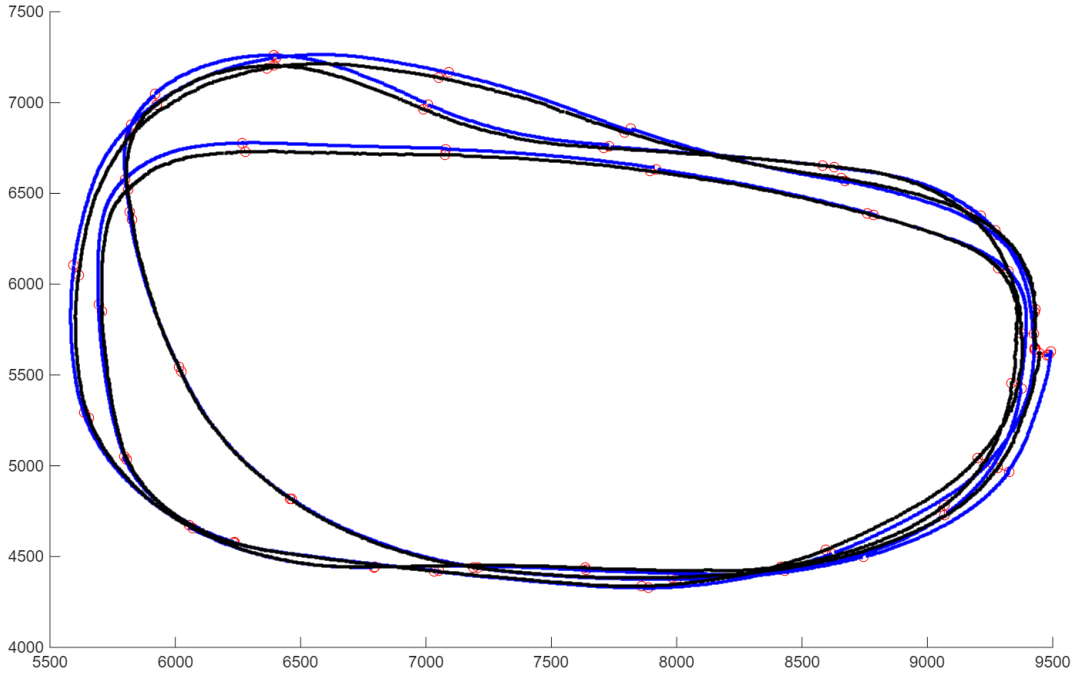


Figure 9: Task 6 path (KF + Cox).

The estimated path (blue) tracks ground truth (black) closely throughout all three loops comparable to Task 4 but with smoother transitions at corrections.

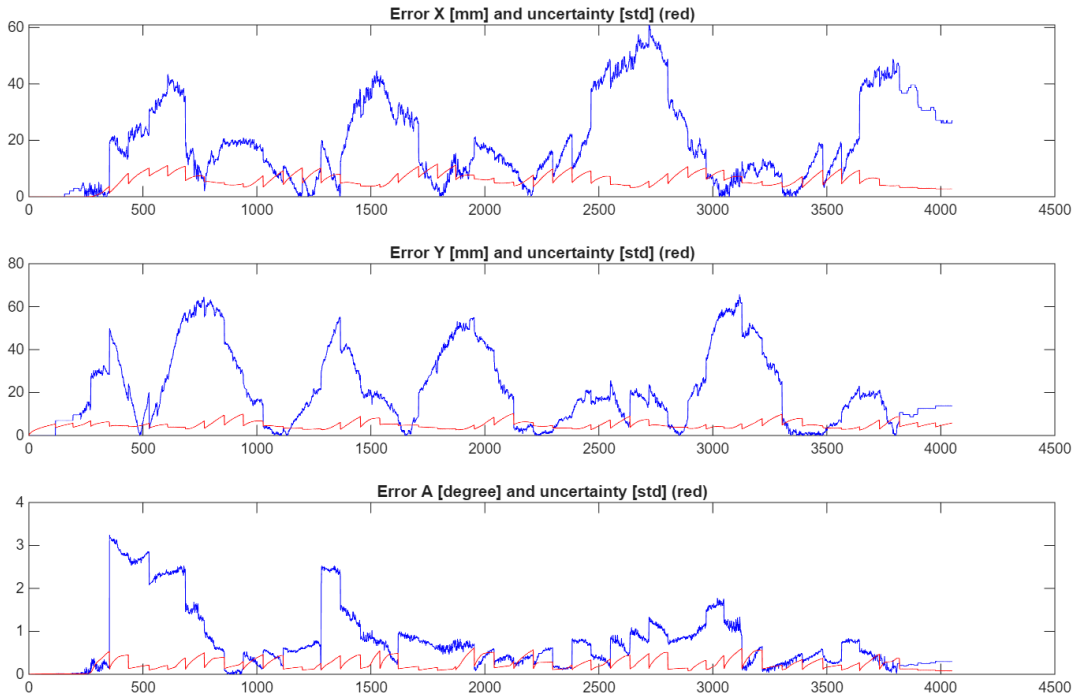


Figure 10: Task 6 errors.

$X$ error peaks at about 60 mm, $Y$ at about 65 mm, heading within about 3°. The $1\sigma$ uncertainty (red) shows periodic growth and reset similar to Task 4.

Unlike Tasks 5a and 5b, the Kalman filter with Cox fixes does not diverge and gives excellent tracking (Figure 9). This is the result of a complete navigation pipeline as dead reckoning provides continuous prediction between scans, the Cox algorithm computes a position fix from the laser data at each scan and the Kalman filter fuses them together weighting each by its covariance.

The reason this works where the simulated fixes failed comes down to the nature of the Cox position fix. The Cox correction $(dx, dy, d\theta)$ is computed relative to the current dead-reckoning estimate so the innovation (difference between fix and prediction) is always small. This avoids the large innovation-rejection problem we saw in Task 5. Also, the Cox covariance $C$ is computed from the actual laser data residuals (Equation (17)) so it reflects the real quality of the scan match rather than some assumed noise level.

The error plot (Figure 10) shows peak errors of about 60 mm in $X$ and 65 mm in $Y$ slightly better than Task 4's 70 mm and 90 mm. The Kalman filter also provides smoother corrections as the Cox match has high uncertainty (poor laser coverage) the filter leans more on dead reckoning and when dead-reckoning uncertainty has grown between scans it trusts the Cox fix more. This adaptive weighting is the main advantage over the direct correction approach in Task 4.

## 3.5 Summary

Table 1 summarizes how the different methods performed.

| Method | Max Error | Error Behavior | Uncertainty |
|---|---|---|---|
| Dead Reckoning (Task 2) | ∼210 mm | Grows without bound | Always increasing |
| Cox only (Task 4) | ∼90 mm | Sawtooth, bounded | Resets at each scan |
| KF + sim. small (5a) | ∼3000 mm | Diverges | Overconfident ($\approx 0$) |
| KF + sim. large (5b) | ∼7000 mm | Diverges (worse) | Overconfident ($\approx 0$) |
| KF + Cox (Task 6) | ∼65 mm | Bounded, smooth | Bounded and adaptive |

Table 1: Comparison of localization methods. Tasks 5a and 5b diverge due to underestimated process noise, while Task 6 succeeds because the Cox covariance is data-driven.

# 4 Conclusions

This exercise gave us hands-on experience with the full sensor-fusion pipeline from dead reckoning, scan matching to Kalman filtering. More than just implementing the algorithms it taught us how each piece fits together.

The most significant learning point was observing the Kalman filter diverge in Tasks 5a and 5b. The filter produced position errors exceeding 3000 mm despite receiving accurate simulated fixes because the process noise model was too optimistic. This demonstrated that the Kalman filter is only optimal when its noise models reflect reality. When the predicted covariance underestimates the true uncertainty the Kalman gain becomes too small and the filter effectively ignores the measurements.

Task 6 confirmed that sensor fusion when properly configured outperforms either source individually. The Kalman filter with Cox fixes achieved about 65 mm maximum error compared to 210 mm for dead reckoning and 90 mm for Cox corrections alone. The key difference from Tasks 5a/5b is that the Cox covariance is computed from the actual scan data which gives the filter a reliable basis for weighting the correction against the prediction.

Implementing the coordinate frame transformations (sensor to robot to world) also reinforced how sensitive the overall pipeline is to correct geometry. An error in any one transformation propagates through the Cox algorithm and produces incorrect line associations which no amount of filtering can recover from.

**Small Suggestions for improvement.** As a minor suggestion, the animated visualization in Exercise 2 was very helpful for building intuition and fun to watch but in Exercises 3 and 4 the figures redraw each iteration so there is no animation to follow and figure 2 (robot moving and scaning) is preety much useless. We found that Python's Matplotlib (`FuncAnimation`) handled this much better when we tried it. We understand that this would add so much more work for the course staff and the student themselves can convert from MatLab to python.

# References


[1] I. J. Cox, "Blanche: An Experiment in Guidance and Navigation of an Autonomous Robot Vehicle," *IEEE Transactions on Robotics and Automation*, vol. 7, no. 2, pp. 193–204, April 1991.

[2] P. S. Maybeck, "The Kalman Filter: An Introduction to Concepts," in *Autonomous Robot Vehicles*, I. J. Cox and G. T. Wilfong, Eds. New York: Springer-Verlag.